Change of Variables in Multiple Integrals, Jacobians

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PRACTISE: HOWARD ANTON: 14.7: Change of Variables in Multiple Integrals, Jacobians: 1-12, 21-24, 35-37

1–4 Find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

1. $x = u + 4v,\ y = 3u - 5v$

2. $x = u + 2v^2,\ y = 2u^2 - v$

3. $x = \sin u + \cos v,\ y = -\cos u + \sin v$

4. $x = \dfrac{2u}{u^2 + v^2},\ y = -\dfrac{2v}{u^2 + v^2}$

5–8 Solve for x and y in terms of u and v, and then find the Jacobian $\partial(x, y)/\partial(u, v)$. ■

5. $u = 2x - 5y,\ v = x + 2y$

6. $u = e^x,\ v = ye^{-x}$

7. $u = x^2 - y^2,\ v = x^2 + y^2 \quad (x > 0, y > 0)$

8. $u = xy,\ v = xy^3 \quad (x > 0, y > 0)$

9–12 Find the Jacobian $\partial(x, y, z)/\partial(u, v, w)$. ■

9. $x = 3u + v,\ y = u - 2w,\ z = v + w$

10. $x = u - uv,\ y = uv - uvw,\ z = uvw$

11. $u = xy,\ v = y,\ w = x + z$

12. $u = x + y + z,\ v = x + y - z,\ w = x - y + z$

21. Use the transformation $u = x - 2y$, $v = 2x + y$ to find

$$\iint_R \frac{x - 2y}{2x + y}\, dA$$

where R is the rectangular region enclosed by the lines $x - 2y = 1$, $x - 2y = 4$, $2x + y = 1$, $2x + y = 3$.

22. Use the transformation $u = x + y$, $v = x - y$ to find

$$\iint_R (x - y)e^{x^2 - y^2}\, dA$$

over the rectangular region R enclosed by the lines $x + y = 0$, $x + y = 1$, $x - y = 1$, $x - y = 4$.

23. Use the transformation $u = \frac{1}{2}(x + y)$, $v = \frac{1}{2}(x - y)$ to find

$$\iint_R \sin \tfrac{1}{2}(x + y) \cos \tfrac{1}{2}(x - y)\, dA$$

over the triangular region R with vertices $(0, 0)$, $(2, 0)$, $(1, 1)$.

24. Use the transformation $u = y/x$, $v = xy$ to find

$$\iint_R xy^3\, dA$$

over the region R in the first quadrant enclosed by $y = x$, $y = 3x$, $xy = 1$, $xy = 4$.

35–38 Evaluate the integral by making an appropriate change of variables. ■

35. $\iint_R \frac{y-4x}{y+4x}\,dA$, where R is the region enclosed by the lines $y = 4x$, $y = 4x + 2$, $y = 2 - 4x$, $y = 5 - 4x$.

36. $\iint_R (x^2 - y^2)\,dA$, where R is the rectangular region enclosed by the lines $y = -x$, $y = 1 - x$, $y = x$, $y = x + 2$.

37. $\iint_R \frac{\sin(x-y)}{\cos(x+y)}\,dA$, where R is the triangular region enclosed by the lines $y = 0$, $y = x$, $x + y = \pi/4$.

38. $\iint_R e^{(y-x)/(y+x)}\,dA$, where R is the region in the first quadrant enclosed by the trapezoid with vertices $(0, 1)$, $(1, 0)$, $(0, 4)$, $(4, 0)$.